The Four Quadrants of Quantum Machine Learning

Concepts, methods, and where the QML Lab project fits

QML Lab project notes

October 25, 2026

Abstract

Quantum machine learning (QML) is not one field but four. A common taxonomy sorts work by two questions: is the data (or the system being studied) classical or quantum, and is the computer that learns from it classical or quantum? The answers give four quadrants, **CC**, **CQ**, **QC** and **QQ**. This document explains each quadrant and the main concepts inside it, with the key equations and references, and then maps the modules and research projects of the QML Lab onto the quadrants. Most of the project lives in **CQ**, the quadrant served by PennyLane and Qiskit; the open-ended projects reach into **QC** and **QQ**, where the strongest current results and the clearest arguments for quantum advantage are found.

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1 Why a taxonomy?

"Quantum machine learning" is used for very different activities: running a neural network on a quantum computer, using a neural network to read out a quantum experiment, borrowing quantum mathematics for classical algorithms, or learning directly from quantum states. These activities need different hardware, rest on different assumptions and make different claims about speed-ups. Lumping them together causes most of the confusion in the field.

The taxonomy used here goes back to Aïmeur, Brassard and Gambs (2006), who distinguished learning settings by whether the learner and the learning task are classical or quantum. It was used in the now-familiar 2×2 form by Dunjko, Taylor and Briegel (2016; 2018) and popularised in the textbook of Schuld and Petruccione (2018; 2021). Broad reviews of the field are refs. (Biamonte et al., 2017; Dunjko and Briegel, 2018; Cerezo et al., 2021a).

In short

Each label has two letters. The *first* letter says what kind of data or system is being studied; the *second* says what kind of computer processes it. **CQ** therefore means *classical data, quantum processor*.

2 Preliminaries: quantum states, measurement and "quantum data"

Qubits and states. A qubit is a unit vector in $\mathbb{C}^2$, $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$ with $|a|^2 + |b|^2 = 1$. A register of n qubits lives in a 2^n-dimensional Hilbert space. This exponential size is the source of both the hopes for QML (large feature spaces) and many of its difficulties (data loading,

vanishing gradients). Mixed states, which describe noise or partial knowledge, are written as density matrices ρ with $\rho \succeq 0$ and $\operatorname{Tr}\rho = 1$.

Measurement. Measuring $|\psi\rangle$ in the computational basis returns outcome i with probability $p(i) = |\langle i|\psi\rangle|^2$ (the Born rule), and the state collapses. What a quantum model outputs in practice is an *expectation value* of an observable O,

$$\langle O\rangle_\rho = \operatorname{Tr}(O\rho), \tag{1}$$

estimated from M repeated measurements (*shots*) with statistical error of order $1/\sqrt{M}$. Every quantum model therefore pays a sampling cost that classical models do not.

Why measurement matters for learning. A measurement extracts at most n classical bits from n qubits and destroys the superposition. Information that lives in phases and correlations of the state, which is most of it, is lost unless the measurement is chosen with the question already in mind. This single fact explains why the QC/QQ distinction matters (section 7).

Quantum data. "Quantum data" means data that *is* a quantum state or process: the output of a quantum simulation, the state stored in a quantum sensor or memory, or a state arriving over a quantum network. It is different from classical data that happens to describe a quantum system (a table of measurement outcomes is classical data).

3 The taxonomy at a glance

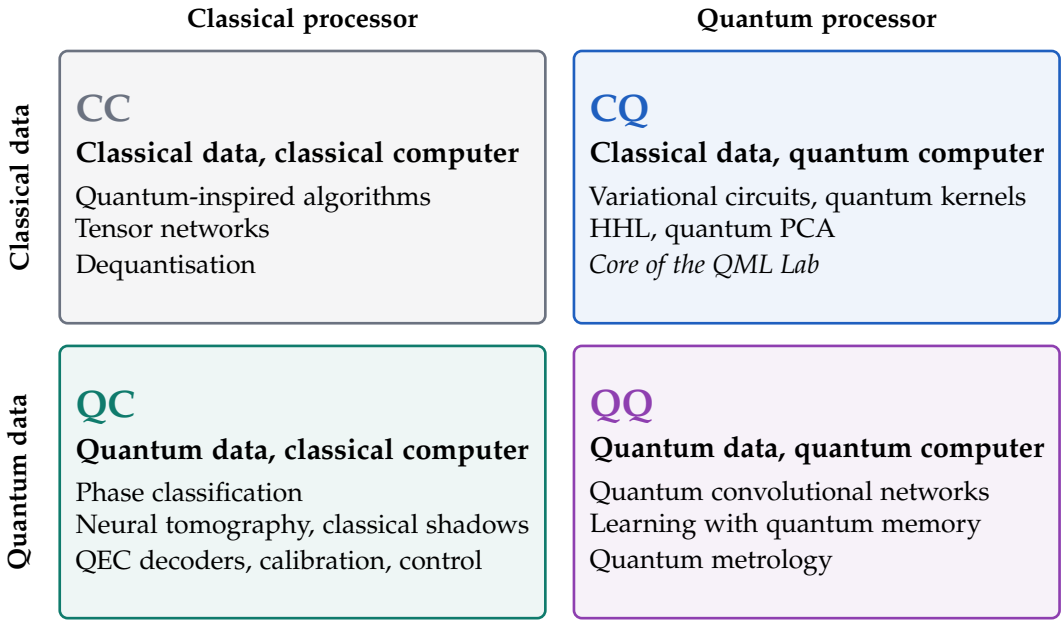


Figure 1: The four quadrants of quantum machine learning. Rows: what is being learned about. Columns: what does the computing.

The two axes separate two questions that are often blurred: *what are we learning about?* and *what are we computing with?* Table 1 summarises the quadrants; the following sections explain each.

	Typical question	Needs quantum hardware?	Status of advantage claims
CC	Can quantum ideas improve classical algorithms?	No	Some proposed quantum speed-ups were matched classically
CQ	Can a quantum computer learn better from ordinary data?	Yes (NISQ or fault-tolerant)	Debated; data loading and trainability are open problems
QC	Can classical ML help us understand or control quantum systems?	Only as the data source	Already useful in practice
QQ	Can a quantum computer learn directly from quantum states?	Yes, plus a quantum data source	Strongest theoretical case; proven separations exist

Table 1: Summary of the four quadrants.

4 CC: classical data, classical processing

At first sight CC is just ordinary machine learning. It belongs in the scheme because of *quantum-inspired* methods: classical algorithms that borrow mathematics or ideas from quantum physics. It also serves a second role: CC is the baseline every quantum claim must beat.

4.1 Tensor networks

Tensor networks were developed to represent many-body quantum states compactly (Orús, 2019). A state of N particles has d^N amplitudes, but physically relevant states often have limited entanglement and can be written as a chain of small tensors, a *matrix product state* (MPS), whose size grows only linearly in N.

Stoudenmire and Schwab (2016) applied the same structure to classification. Each input feature $x_j \in [0, 1]$ is mapped to a two-component vector, and the features are combined with a tensor product:

$$\phi(x_j) = \begin{pmatrix} \cos \frac{\pi}{2} x_j \\ \sin \frac{\pi}{2} x_j \end{pmatrix}, \qquad \Phi(x) = \phi(x_1) \otimes \phi(x_2) \otimes \cdots \otimes \phi(x_N), \qquad f^\ell(x) = W^\ell \cdot \Phi(x). \tag{2}$$

$\Phi(x)$ lives in a 2^N-dimensional space, like an N-qubit product state. The weight tensor W^ℓ for class ℓ is stored as an MPS with bond dimension χ, so evaluation and training cost grows only linearly in N and polynomially in χ. The bond dimension controls how much "entanglement", that is how many feature correlations, the model can represent.

4.2 Dequantisation

Several early CQ algorithms promised exponential speed-ups for linear-algebra tasks. Kerenidis and Prakash (2017) proposed a quantum recommendation system whose running time is polylogarithmic in the size of the preference matrix. Its input model assumes the data can be prepared as quantum states efficiently, for example from a quantum random access memory (QRAM) (Giovannetti et al., 2008).

Tang (2019) showed that a *classical* algorithm given the analogous access, namely the ability to sample rows and entries with probability proportional to their squared norm (ℓ_2-*norm sampling*),

can also produce recommendations in time polylogarithmic in the matrix dimensions, for low-rank matrices. The exponential gap disappears once both sides get the same input assumptions. Chia et al. (2022) turned this into a general framework that dequantises a broad class of low-rank quantum linear-algebra algorithms.

4.3 Why CC matters for the rest

- Speed-up claims are only meaningful against the *best* classical method with the *same* access to the data. Dequantisation is the clearest example.
- Every **CQ** experiment needs a CC baseline with a comparable number of parameters, tuned with equal care (Bowles et al., 2024).

In short

CC is classical computing with quantum mathematics. Tensor networks give compact models; dequantisation shows that some promised quantum speed-ups came from input assumptions rather than from quantum mechanics.

5 CQ: classical data, quantum processing

This is what most people mean by quantum machine learning and what most PennyLane and Qiskit tutorials teach: classical data (images, tables, time series) is encoded into qubits and processed by a quantum circuit. A **CQ** model has three stages,

$$x \xrightarrow{\text{encode}} |\psi(x)\rangle \xrightarrow{\text{process}} U(\theta)\,|\psi(x)\rangle \xrightarrow{\text{measure}} \langle O\rangle,$$

and usually a classical optimiser that adjusts θ.

5.1 Data encoding (feature maps)

The *feature map* $x \mapsto |\psi(x)\rangle$ decides what the model can learn, because the trainable part can only separate what the encoding has already pulled apart (Schuld and Killoran, 2019). Common choices:

Basis encoding
A bit string $x \in \{0,1\}^n$ becomes the basis state $|x\rangle$. Natural for binary data; one qubit per bit.

Angle encoding
Each feature is a rotation angle, $|\psi(x)\rangle = \bigotimes_{j=1}^{n} R_Y(x_j)\,|0\rangle$. One qubit per feature, cheap to prepare, but a product state with no entanglement between features.

Amplitude encoding
A normalised vector becomes the amplitudes,

$$|\psi(x)\rangle = \frac{1}{\|x\|}\sum_{i=1}^{N} x_i\,|i\rangle\,, \tag{3}$$

using only $\lceil \log_2 N \rceil$ qubits. Very compact, but preparing a general state needs circuits of size $O(N)$, and the norm $\|x\|$ is lost.

IQP / ZZ feature maps
Hadamards followed by diagonal gates whose phases depend on single features and on products of features (Havlíček et al., 2019):

$$U_\Phi(x) = \exp\Big(i \sum_S \phi_S(x) \prod_{j\in S} Z_j\Big) H^{\otimes n}, \qquad \phi_{\{i\}}(x) = x_i, \quad \phi_{\{i,j\}}(x) = (\pi - x_i)(\pi - x_j), \tag{4}$$

usually applied more than once. These states are believed to be hard to simulate classically, which is what makes the corresponding kernel (section 5.6) interesting.

5.2 The data-loading problem

Many theoretical **CQ** speed-ups assume that a data vector can be turned into a quantum state in time polylogarithmic in its length. For arbitrary classical data that needs a QRAM (Giovannetti et al., 2008), which does not exist at scale, or a state-preparation circuit whose cost can be linear in the data size and cancel the speed-up. Aaronson (2015) lists this as the first item of "fine print" to check in any quantum learning claim, together with the condition number of the problem, the fact that the output is a quantum state rather than a list of numbers, and comparison with the best classical algorithm.

5.3 Variational quantum circuits

A variational quantum circuit (also called a parameterised quantum circuit or quantum neural network) defines the model

$$f_\theta(x) = \langle \psi(x) | U(\theta)^\dagger \, O \, U(\theta) | \psi(x) \rangle \,, \tag{5}$$

where $U(\theta)$ is a sequence of parameterised rotations and entangling gates (the *ansatz*) and O an observable such as Z on the first qubit (Mitarai et al., 2018; Cerezo et al., 2021a). For classification, the sign of $f_\theta(x)$ plus a bias gives the label. Training minimises a loss such as the mean squared error with a classical optimiser. Only the circuit evaluations run on quantum hardware, which makes these models suitable for noisy intermediate-scale quantum (NISQ) devices (Preskill, 2018).

5.4 Gradients: the parameter-shift rule

Back-propagation is impossible on hardware, because intermediate states cannot be stored or copied. For gates of the form $e^{-i\theta P/2}$ with P a Pauli operator, the exact derivative follows from two additional circuit runs (Mitarai et al., 2018; Schuld et al., 2019):

$$\frac{\partial f}{\partial \theta_k} = \frac{1}{2}\Big[f\big(\theta + \tfrac{\pi}{2} e_k\big) - f\big(\theta - \tfrac{\pi}{2} e_k\big) \Big]. \tag{6}$$

Unlike finite differences this is exact, and it works on hardware. Its cost is two circuit evaluations per parameter per step, each estimated from shots.

5.5 Expressivity and data re-uploading

If a feature enters the circuit through gates e^{-ixG}, the model is a truncated Fourier series in x (Schuld et al., 2021),

$$f_\theta(x) = \sum_{\omega \in \Omega} c_\omega(\theta) \, e^{i\omega x}, \tag{7}$$

where the available frequencies Ω are set by the eigenvalues of the encoding generators. Encoding a feature once with a single Pauli rotation gives only frequencies $\{-1, 0, 1\}$: a single sine wave. *Data re-uploading* (Pérez-Salinas et al., 2020), encoding x again before each trainable layer, enlarges the spectrum to $\{-L, \dots, L\}$ for L repetitions. This is why even one or two qubits can learn curved decision boundaries when the data is re-uploaded.

5.6 Quantum kernel methods

A kernel method needs only a similarity measure between data points. A quantum kernel uses the overlap of encoded states (Havlíček et al., 2019; Schuld and Killoran, 2019):

$$\kappa(x, x') = |\langle \psi(x) | \psi(x') \rangle|^2 = \mathrm{Tr}\big[\rho(x)\,\rho(x')\big]. \tag{8}$$

The quantum computer only estimates the Gram matrix $K_{ij} = \kappa(x_i, x_j)$, for example by running $U_\Phi(x')^\dagger U_\Phi(x)$ and counting how often all qubits return to $|0 \cdots 0\rangle$. A classical support vector machine then learns the decision function $f(x) = \mathrm{sign}\left(\sum_i \alpha_i y_i \kappa(x_i, x) + b\right)$. Training is a convex problem, so there are no barren plateaus in the optimisation itself. Variational models and kernel models are closely related: the model in eq. (5) is a linear model in the feature space of density matrices (Schuld and Petruccione, 2021).

The catch is *exponential concentration* (Thanasilp et al., 2024). For expressive feature maps on many qubits, the off-diagonal kernel values concentrate exponentially around a fixed value, so the Gram matrix approaches the identity and an exponential number of shots is needed to tell entries apart. The resulting model memorises the training data and generalises poorly. Huang et al. (2021) add a second caution: given enough data, classical models can often match quantum kernels, and they give a test (the geometric difference between kernels) for when a quantum kernel can possibly help.

5.7 Quantum linear-algebra algorithms

A separate line of **CQ** work speeds up the linear algebra underneath many learning algorithms. Such algorithms usually assume fault-tolerant hardware.

- **HHL** (Harrow et al., 2009) prepares a state $|x\rangle \propto A^{-1}|b\rangle$ for a sparse, well-conditioned $N \times N$ matrix A in time $\tilde{O}(\log(N)\, s^2 \kappa^2/\epsilon)$, where s is the sparsity and κ the condition number. The output is a quantum state; reading out all of x would cost $O(N)$ again.
- **Quantum PCA** (Lloyd et al., 2014) uses copies of a density matrix ρ to apply $e^{-i\rho t}$ and phase estimation to reveal the largest eigenvalues and eigenvectors of ρ in time logarithmic in its dimension, for low-rank ρ.
- **Quantum SVM** (Rebentrost et al., 2014) solves the least-squares SVM with these primitives.

Each of these inherits the caveats of section 5.2, and several low-rank variants have been dequantised (section 4).

5.8 Trainability and noise

Barren plateaus

For deep, randomly initialised circuits the gradient of the cost averages to zero, and its variance vanishes exponentially in the number of qubits n (McClean et al., 2018): $\mathrm{Var}[\partial_k C] \in O(b^{-n})$ for some $b > 1$. The landscape is flat almost everywhere. Global cost functions

(observables acting on all qubits) suffer even for shallow circuits, while local costs on shallow circuits keep polynomially large gradients (Cerezo et al., 2021b).

Noise
Gate errors and decoherence shrink expectation values toward their maximally mixed values, so deeper circuits carry less signal. On NISQ hardware, circuits must stay shallow.

Shot noise
Every expectation value, gradient and kernel entry is estimated from finitely many shots, so small signals cost many repetitions.

In short

CQ is the most explored quadrant and the one the QML Lab modules teach. Its tools are feature maps, variational circuits trained with the parameter-shift rule, and quantum kernels. Its open problems are data loading, barren plateaus, kernel concentration, noise, and showing an advantage over strong classical baselines.

6 QC: quantum data, classical processing

Here classical machine learning is the tool and quantum systems are the subject. The data comes from quantum experiments or simulations, but the learning runs on ordinary computers. This quadrant is already productive because it needs no fault-tolerant quantum computer (Gebhart et al., 2023).

6.1 Classifying phases of matter

Carrasquilla and Melko (2017) trained ordinary neural networks on spin configurations sampled from many-body models. The networks learned to distinguish ordered from disordered phases and located phase transitions without being told the order parameter. The input is measurement data, that is classical snapshots of quantum states, and the processor is classical.

6.2 Neural-network quantum states and tomography

A neural network can represent the wave function itself. Carleo and Troyer (2017) used a restricted Boltzmann machine,

$$\psi_\theta(\sigma) = \sum_{\{h\}} \exp\Big(\sum_j a_j \sigma_j + \sum_i b_i h_i + \sum_{ij} W_{ij} h_i \sigma_j \Big), \tag{9}$$

to find ground states of spin models variationally. Torlai et al. (2018) turned this around for *tomography*: training such a network on measurement outcomes reconstructs a many-body state from far fewer measurements than full tomography, whose cost grows exponentially with system size.

6.3 Classical shadows

Huang, Kueng and Preskill (2020) showed how to predict many properties of a state from few measurements. Each round applies a random unitary U (for example random single-qubit

Pauli bases), measures in the computational basis, and stores a classical "snapshot",

$$\hat{\rho} = \bigotimes_{j=1}^{n} \left(3\, U_j^\dagger\, |\hat{b}_j\rangle\langle\hat{b}_j|\, U_j - I\right). \tag{10}$$

Averages of snapshots predict M observables to accuracy ϵ with a number of measurements that grows only as $\log M$ (for local observables). The measurement is quantum, the learning is classical: a clean example of **QC**, and a common input format for classical ML on quantum data.

6.4 Decoders for quantum error correction

An error-correcting code reports a *syndrome*, a pattern of parity checks, from which a decoder must infer the most likely error. Torlai and Melko (2017) trained a neural network to do this for topological codes. Fast, accurate decoders are a bottleneck for fault-tolerant hardware, and machine-learned decoders are an active area.

6.5 Calibration, control and characterisation

Machine learning tunes the hardware itself: optimising control pulses, calibrating gates, and estimating noise. Bukov et al. (2018) used reinforcement learning to find control protocols for preparing target states. The review by Gebhart et al. (2023) surveys learning for characterisation, verification and control.

In short

QC uses classical ML to understand and control quantum systems: phase detection, neural-network states and tomography, classical shadows, decoders and calibration. It works today, at scale, because the learner is classical.

7 QQ: quantum data, quantum processing

In **QQ** the data is already quantum, and a quantum computer processes it directly without first measuring it into classical numbers. This avoids the information loss of measurement (section 2) and the data-loading problem of section 5.2, because nothing needs to be loaded.

7.1 Quantum convolutional neural networks

Cong, Choi and Lukin (2019) proposed a circuit with the structure of a convolutional network. *Convolution* layers apply the same quasi-local unitary across the register; *pooling* layers measure some qubits and use the results to rotate their neighbours, halving the system size; a final fully connected unitary acts on the few remaining qubits. For N input qubits a QCNN has only $O(\log N)$ variational parameters. It was shown to recognise symmetry-protected topological phases directly from the quantum state, and its structure helps avoid barren plateaus.

7.2 Quantum autoencoders

Romero, Olson and Aspuru-Guzik (2017) train an encoder $U(\theta)$ that compresses a family of n-qubit states into $n-k$ qubits. The cost asks the k discarded ("trash") qubits to end up in a

fixed reference state; if they do, the state can be decoded with $U(\theta)^\dagger$ without loss. Applications include compressing the output of quantum simulations and denoising.

7.3 Learning with quantum memory

The strongest results in **QQ** concern *learning from experiments*. A conventional learner measures each copy of an unknown state or each use of an unknown process and learns from the classical outcomes. A quantum-enhanced learner stores copies in quantum memory and processes them jointly, for example with entangled measurements across two copies. Huang et al. (2022) proved that for several tasks, such as predicting properties of physical systems and learning about unknown quantum processes, conventional learners need exponentially more experiments than quantum-enhanced ones, and demonstrated the effect on a superconducting processor. This is a provable, experimentally tested separation, the clearest one in QML so far.

7.4 Quantum metrology

Estimating an unknown parameter φ (a phase, a field, a frequency) is a learning task on quantum data. With N independent probes, the error scales as the standard quantum limit $\Delta\varphi \propto 1/\sqrt{N}$; entangled probes and quantum processing can reach the Heisenberg limit $\Delta\varphi \propto 1/N$ (Giovannetti et al., 2011). Adaptive, learning-based estimation strategies connect metrology with machine learning (Gebhart et al., 2023).

7.5 What QQ needs

QQ needs a quantum data source *connected* to a quantum processor: a quantum sensor, a quantum simulator or a quantum network link feeding qubits into a processor without measuring them. Such pipelines exist only in early experiments, which is why **QQ** is the most promising quadrant in theory and the least developed in practice.

In short

QQ processes quantum states without measuring them first. It has the clearest theoretical case for quantum advantage, including proven exponential separations in learning from experiments, but it needs quantum data sources wired directly to quantum processors.

8 Comparing the quadrants

Where does quantum advantage come from? **CC** and **QC** use classical computers, so they work at scale today. In **CQ** the case for advantage rests on feature spaces or linear algebra that are hard to simulate classically, but it is weakened by data loading, dequantisation and the trainability problems above. In **QQ** the advantage has a simple source: measuring destroys information, and a quantum processor can avoid measuring.

Is advantage the right goal? Schuld and Killoran (2022) argue that chasing speed-up claims has distorted the field, and that QML should also be studied for what it teaches about learning, quantum models and their inductive biases. Bowles, Ahmed and Schuld (2024) benchmarked many published quantum models on standard tasks and found that well-tuned classical models usually matched or beat them, and that removing entanglement often did not hurt. Careful benchmarking is part of doing QML well.

	Key concepts	Main obstacle	Representative work
CC	tensor networks, ℓ_2-norm sampling, dequantisation	none quantum; must beat strong baselines	Stoudenmire and Schwab, 2016; Tang, 2019; Chia et al., 2022
CQ	feature maps, variational circuits, parameter shift, kernels, HHL	data loading, barren plateaus, concentration, noise	Havlíček et al., 2019; Schuld and Killoran, 2019; Mitarai et al., 2018; Harrow et al., 2009; McClean et al., 2018
QC	phase classification, neural quantum states, classical shadows, decoders	measurement cost, model expressivity	Carrasquilla and Melko, 2017; Carleo and Troyer, 2017; Torlai et al., 2018; Huang et al., 2020
QQ	QCNNs, autoencoders, quantum memory, metrology	connecting quantum data sources to processors	Cong et al., 2019; Romero et al., 2017; Huang et al., 2022; Giovannetti et al., 2011

Table 2: Concepts, obstacles and key references per quadrant.

9 Where the QML Lab project fits

The QML Lab is a ten-week study plan (modules 1–8) built on PennyLane (Bergholm et al., 2018) and Qiskit (Javadi-Abhari et al., 2024), plus sixteen open-ended research projects managed in a Django portal. Its base is **CQ**: the modules encode classical data into qubits and train circuits and kernels on it. Every project also carries a classical baseline, which is CC by design.

Module	Topic	Quadrant
1–3	Qubits, gates, entanglement, parameter-shift gradients	foundations
4	Data encoding: angle, amplitude, ZZ feature maps	**CQ**
5	Variational classifier with data re-uploading	**CQ**
6	Quantum kernels and SVM, with an RBF baseline	**CQ** (+ CC baseline)
7	Hybrid quantum–classical network in PyTorch	**CQ** (+ CC baseline)
8	Barren plateaus and noise	**CQ** trainability

Table 3: The learning modules and their quadrant.

Each project has its own living document (`qml_projects/<slug>/docs/main.tex`) with background, research questions, method, planned experiments and a results log, using the same Harvard-style bibliography as this overview.

Reading the mapping. Twelve of the sixteen projects are **CQ**, matching the tools and simulators available. For a thesis with a stronger theoretical case for quantum advantage, projects 6 and 12 (**QQ**) are the natural direction; project 13 (**QC**) is the most immediately useful for work on real hardware. Every project follows the same quality bar: a classical baseline of similar size, several random seeds, scaling with qubit count, and at least one run under a realistic noise model (Bowles et al., 2024).

#	Project	Quadrant and reason
1	Quantum-kernel intrusion detection	CQ: network-traffic features, quantum kernel
2	Trainable quantum kernels	CQ: learned feature map
3	Projected vs fidelity kernels	CQ; borders CC through classical simulability
4	Re-uploading depth study on MNIST	CQ
5	QAOA for MaxCut with learned angles	QC-like: a classical model learns to set up a quantum algorithm (Farhi et al., 2014)
6	Quantum autoencoder	QQ: compresses quantum states
7	Quantum GAN for a 1-D distribution	CQ
8	Hybrid modulation classifier	CQ
9	Quantum transfer learning	CQ
10	Quantum reinforcement learning on CartPole	CQ
11	Quantum reservoir computing	CQ
12	QCNN for phase classification	QQ: quantum ground states in, quantum processing
13	Learning from classical shadows	QC: random measurements, classical estimation
14	Noise-aware training with error mitigation	CQ under realistic noise
15	Barren-plateau mitigation	CQ trainability
16	Simulator vs IBM hardware benchmark	CQ on real hardware

Table 4: The sixteen research projects by quadrant.

10 Glossary

Ansatz
The fixed structure of a parameterised circuit; only its angles are trained.

Barren plateau
A cost landscape whose gradients vanish exponentially with qubit number.

Bond dimension
The size of the indices linking tensors in a tensor network; it bounds the correlations the network can represent.

Classical shadow
A classical record of randomised measurements used to predict many properties of a quantum state.

Data re-uploading
Encoding the same input several times between trainable layers to make the model more expressive.

Dequantisation
A classical algorithm that matches a claimed quantum speed-up under the same input assumptions.

Exponential concentration
Kernel values or expectation values collapsing toward a constant as qubits are added, so that exponentially many shots are needed to resolve them.

Feature map
The map $x \mapsto |\psi(x)\rangle$ from data to quantum states.

NISQ
Noisy intermediate-scale quantum: today's devices with tens to hundreds of noisy qubits and no full error correction (Preskill, 2018).

Parameter-shift rule
Exact gradients of circuit outputs from two shifted circuit runs.

QRAM
Quantum random access memory: a hypothetical device that loads classical data into superposition efficiently.

Quantum data
Data that is a quantum state or process, not a classical description of one.

Shot
One execution and measurement of a circuit; expectation values are averages over shots.

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